Remanent Magnetism in Jurassic Red Limestones and Radiolarites from the Alps

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Summary

Red Jurassic limestones and radiolarites from the Kammerköhr range on the Austrian–German frontier show rather consistent remanent magnetic vectors. The average North Pole position for lower Jurassic rocks (red limestones) is latitude 53° N, longitude 112° E and for Middle Jurassic rocks (red radiolarites) latitude 58°·5 N, longitude 128° E.

1. Introduction

In the Northern Calcareous Alps the major portion of the Jurassic system is represented by a rather thin section (of the order of 100–200 m) of fine-grained marine rocks, varying from grey and red limestones (the Kieselknollenkalk, red cephalopod limestone, red Knollenkalk, Adneth beds, etc.) to maroon or greenish grey radiolarites—impure, calcareous cherty rocks replete with the remains of radiolarian tests.

In 1957 Fischer measured a detailed section of this sequence, and collected oriented samples for magnetic analysis. This was carried out by Hargraves, with a "spinner"-type magnetometer, kindly made available by the Geophysical Laboratory of the United States Geological Survey, through the courtesy of Dr James R. Balsley.

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2. Location

The profile sampled is exposed on the west face of the Scheibelberg in the Kammerköhr range. It lies essentially on the German–Austrian frontier, and very near the boundary between the Austrian provinces of Salzburg and Tirol. The co-ordinates are 47° 37′·5 N, 12° 34′·5 E.

The structure of the area is rather simple. The rocks sampled are flat-lying, from the axis of an open syncline which plunges gently eastward, to vanish at Lofer, some 10 km to the east, under the Berchtesgaden thrust sheet. There is some question as to whether this syncline and the adjacent rocks are essentially in place, or transported as part of a thrust sheet. Various thrusts have been mapped farther north, along the Alpine margin, and by analogy with the western Alps various geologists have believed these to be essentially flat. The adherents of this view assume that one or more of these thrusts are present at a depth below the

Kammerköhr range, and that the range is part of a "Staufen-Höllengebirge thrust sheet" (Schaffer 1951). Others have suggested that the frontal thrusts of the eastern Alps are high-angle structures, and that the rocks behind them are essentially autochthonous. In any case, few would care to defend the ultranappist view that these rocks originated on the southern margin of the Alpine trough. If they were thrust, movement was probably of the order of kilometres or tens of kilometres. It has therefore not moved the rocks over a sufficient distance to invalidate their palaeomagnetic applications; neither can one visualize tectonic rotation sufficiently great to have disoriented the magnetic vectors.

3. Stratigraphy and sedimentation

The general stratigraphy of the northern Alps is described in many standard works, among them Schaffer (1951). That of the Kammerköhr range was described by Hahn (1910). Here the Lower Jurassic is represented by limestones—red limestones of various types which interfinger northwestward with the grey to white, cherty Kieselknollenkalk. Most of the standard Liassic ammonite zones have been recognized within this sequence. These limestones are generally conformably overlain by radiolarites—bedded, impure, calcareous, radiolarian cherts, mainly of maroon colour, which in turn grade upward, and in their upper part interfinger with the siliceous Oberalm limestones of latest Jurassic (Tithonian) age. Diagnostic fossils are lacking in the radiolarite, but its stratigraphic relations suggest that the 20 m or so of radiolarite represent essentially all of the Middle and most of the Upper Jurassic.

The profile measured by Fischer on the Scheibelberg is shown in Figure 1. Here the Kieselknollenkalk represents the lower and middle portions of the Lower Jurassic. Hahn (1910) reports *Aegoceras*, an indication that the unit extends into the lower Pliensbachian.

The overlying red limestones and mudstones carry *Amaltheus* (Hahn 1910) and *Harpoceras*, and represent the upper part of the Lias—specifically, the middle and upper Pliensbachian and the lower Toarcian. The slumped zone and succeeding break at the top of this section suggest that the upper Toarcian is not represented.

The radiolarite here is about 20 m thick, but only the lower 15 m are shown in Figure 1. The basal portion is green to grey. The upper contact with the Oberalm beds is covered.

The Kieselknollenkalk and the basal, grey portion of the radiolarite are not sufficiently magnetized to permit determinations of remanent magnetism. The samples here reported on—shown at the left side of Figure 1—are in two distinct groups: One from the red limestones (late Lower Jurassic), and one from the maroon radiolarite. It appears probable that samples 170 up to 369 are Middle Jurassic; sample 381 might be Upper Jurassic, but in the absence of recognizable guide fossils the Middle–Upper Jurassic boundary within the radiolarite remains undefined. The radiolarite samples were therefore treated as one unit.

The Jurassic sediments are quite unlike those found in normal epicontinental deposits. Their fine grain, pelagic faunas, and highly oxidized state are more suggestive of modern deep-sea deposits, and suggest deposition in a "starved" basin, in which subsidence was more rapid than sedimentation. The uniformity and apparent slowness of deposition are in harmony with this interpretation. If the section is as complete as stratigraphic relations suggest, rates of sedimentation for

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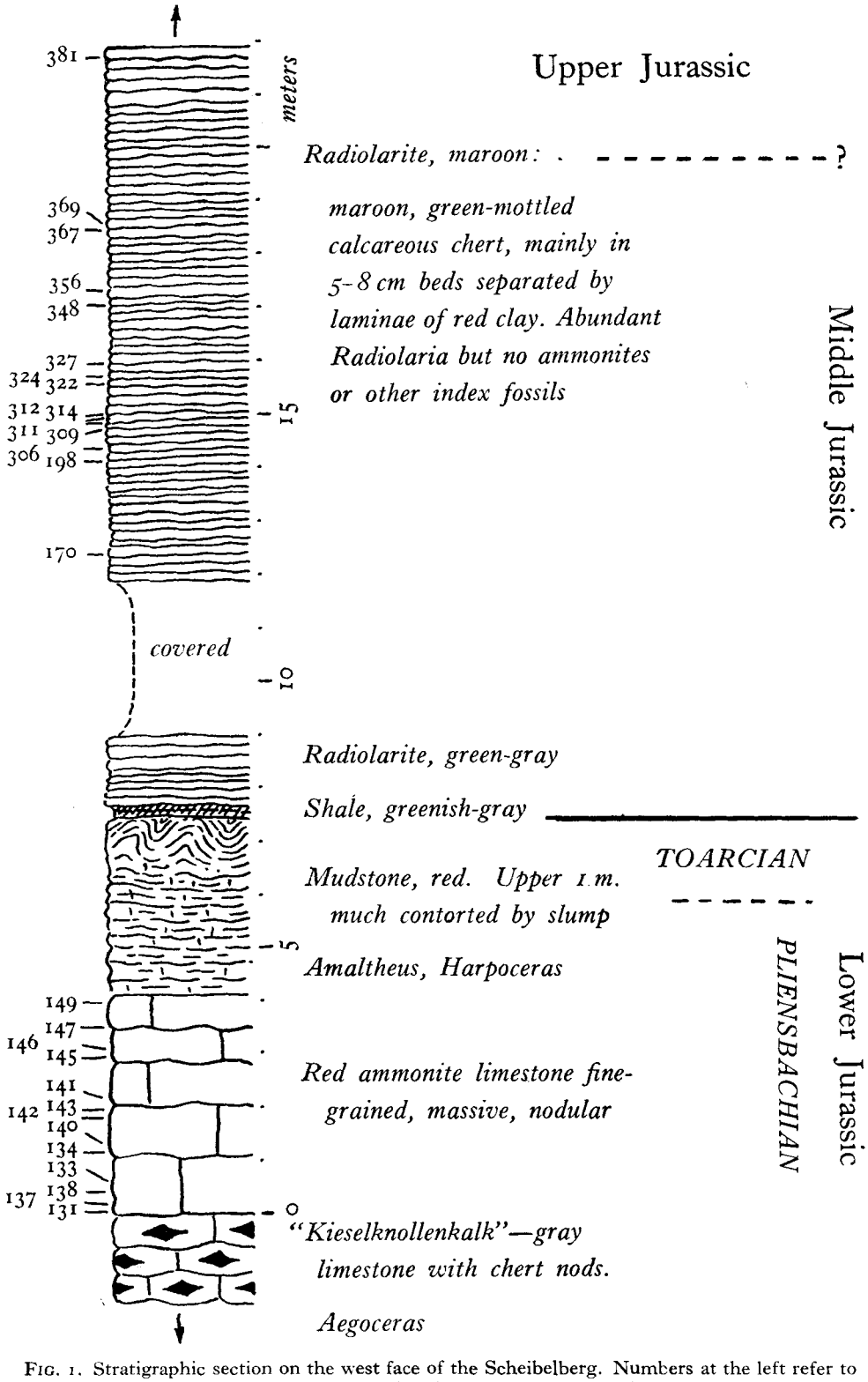


FIG. 1. Stratigraphic section on the west face of the Scheibelberg. Numbers at the left refer to oriented samples analyzed for remanent magnetism.

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the red limestone were of the order of a hundred thousand years per metre, those for the radiolarite of the order of a million years per metre.

4. Remanent magnetism

The remanent magnetism of these red rocks is weak, the limestone averaging $4{\cdot}4 \times 10^{-7}$ emu/cm^3, the radiolarite, $10{\cdot}4 \times 10^{-7}$ emu/cm^3. The direction of magnetization is normal, pointing to a North Pole in the northern hemisphere. The measurements are summarized in Tables 1 and 2, and in Figures 2 and 3. They fall into two groups—one containing the Lower Jurassic limestones, the other the Middle Jurassic radiolarites—which overlap by about 50 per cent of their spread.

Within each group there is a distinct clustering of points in the mid-section. Otherwise, distribution of readings within the groups is random, bearing no relation to the detailed position of cores within the limestone or within the radiolarite. Differences in readings on twin cores cut from the same field sample range up to between a quarter and a third of the total spread. Pole positions, as determined by the average for each group, are shown in Table 2, and lie as follows: Lower Jurassic pole east of Lake Baikal; Middle Jurassic pole in the Aldan region, south of Yakutsk.

5. Discussion of pole positions

The magnetic results obtained are remarkably consistent when compared with the scatter generally obtained in the study of red siltstones and lavas. In view of the short-term gyrations of the pole and of a series of introduced errors—from the field orientation of samples to the cutting in the laboratory and to errors in final measurement—a greater consistency is hardly to be expected. Our samples show none of the magnetic instability observed by Creer (1957) in fine-grained red Keuper marls; on the contrary, they show that some fine-grained haematite-bearing, magnetite-free red sediments constitute exceptionally suitable subjects for palaeomagnetic investigations.

The pattern of polar wandering indicated by European palaeomagnetic data, enabled Irving (1958) to interpolate a Jurassic pole between Triassic and Eocene poles determined from Britain. The position of this interpolated Jurassic pole is about latitude 60° N, longitude 131° E. Our mid-Jurassic pole determination of latitude 58°·5 N, longitude 128° E, agrees closely with this prediction.

6. Origin of remanent magnetism

X-ray diffraction studies of the radiolarite and of insoluble residues of the limestone, indicate that haematite is the only potentially ferromagnetic mineral present. The haematite occurs as an extremely fine-grained red pigment; all attempts to concentrate it by physical methods were unsuccessful.

Three hypotheses may be advanced to account for the origin of the haematite and thereby the remanent magnetism shown by these rocks: (1) *primary*, (2) *penecontemporaneous* (early diagenetic), or (3) *epigenetic*.

(1) *Primary*. According to this hypothesis, the haematite particles originated in lateritic soils on the mainland or on islands within the Tethyan seaway, were carried by air or water currents and were deposited mechanically. Possessing a stable remanent magnetism acquired in or inherited from the source area, the

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Table 1.

Magnetic Vectors of Samples

Stratigraphic position of the samples is shown in Figure 1. A, B and C denote distinct field samples taken in close proximity; a and b denote twin cores taken from the same field sample.

Sample No.	Azimuth	Inclination	
381a	50	37	
b	49	39	
369a	22	50	
b	25	50	
367	22	56	
356a	34	49	
b	32	52	
348a	15	47	
b	17	41	
327	42	49	
324a	27	51	
b	25	46	
322	28	43	
314	42	48	
312a	38	45	↑ Radiolarite
b	40	50	
311	26	49	
309	37	38	
306	32	49	
198	32	49	
170	46	58	
149a	60	54	
b	65	63	
147	56	60	
146a	38	56	
b	38	52	
145a	29	49	
b	29	47	
143a	44	56	↓ Limestone
b	48	43	
142	27	56	
141	49	31	
140C	32	59	
140Ba	43	51	
b	40	53	
140Aa	41	48	
b	37	47	
134a	58	59	
b	61	52	
133a	42	45	
b	49	56	
138Ba	46	42	
b	34	39	
138Aa	44	41	
b	44	42	
137a	44	50	
b	45	45	
131a	53	55	
b	47	53	
120a	44	50	
b	41	42	

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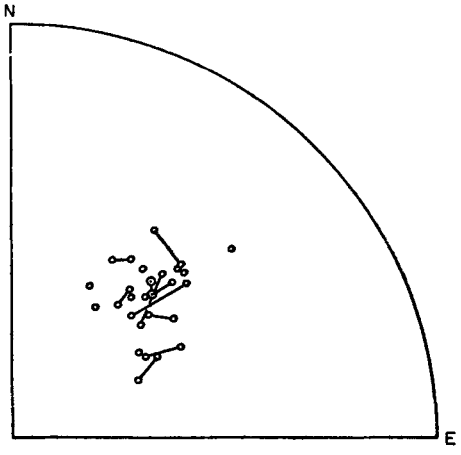


FIG. 2. Equal-area net plot of Mid- Jurassic (radiolarite) samples.

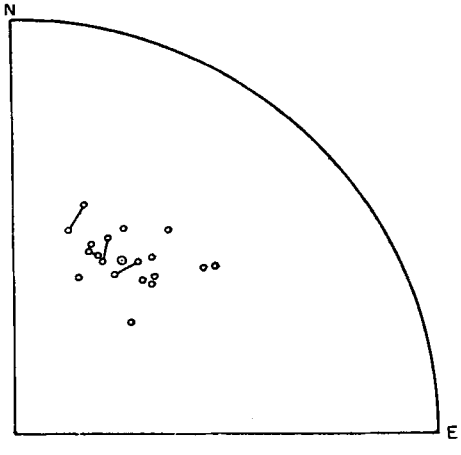


FIG. 3. Equal-area net plot of Lower Jurassic (limestone) samples.

○ Magnetic reading of one core.

⊙ Two superimposed readings from different cores.

Twin cores cut from the same field sample.

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Table 2

Summary of Magnetic Results.

Point of observation: latitude 47°37′ N, longitude 12°34′ E.

	N.R.M.			Semi angle of cone of 95 per cent confidence	Pole position	
	Average azimuth	Average inclination	k of K		Lat. N	Long. E
Radiolarite[1] (Mid-Jurassic)	36°·7	47°·8	100	5°·5	58°·5	128°
Limestone[2] (Lower Jurassic: Pliensbachian)	47°·9	50°·6	70·7	6°·5	53°	112°

(1) based on 21 cores from 15 field samples
(2) based on 30 cores from 16 field samples
Statistics calculated according to procedure recommended by Fisher (1953), Watson & Irving (1957).

particles became oriented in the Earth's magnetic field while settling and thus imparted a primary magnetic fabric to the sediment.

(2) *Penecontemporaneous or early diagenetic.* Another possibility is that the iron reached the site of deposition in solution and was there chemically precipitated. The haematite particles could have formed directly, or been developed during the process of diagenesis (e.g. dehydration of goethite). In either case the remanent magnetism, acquired *pari passu* with sedimentation, would be of chemical origin, the result of crystallization within the Earth's field. The existence of such a "chemical remanent magnetism" has been suggested by Martinez (1956), Howell & others (1958), and demonstrated by Nagata (1958).

(3) *Epigenetic.* A third possibility is that the haematite may be of late diagenetic origin, with an equivalent "epigenetic" magnetization. Alternatively, a haematitic sediment of either primary or penecontemporaneous origin may have received a secondary magnetic overprint at some later time as a result of metamorphism, electrical discharge, or some other undetermined factor.

In the case of the rocks studied, epigenetic magnetization appears the least likely of the three possibilities; haematite is magnetically highly stable; the rocks show no signs of metamorphism, no trace of flowage or cleavage. They were quite possibly overridden by the Berchtesgaden thrust sheet, but may never have been buried to depths greater than 3 or 4 thousand metres. Moreover, epigenetic magnetization of late diagenetic origin, or in response to temperature changes or orogenic events would most likely have occurred during late Cretaceous or Tertiary time, and should in that case have imparted a magnetism more nearly like the Earth's present field than that observed.

It therefore appears that the magnetism encountered is either a primary (physical) or early diagenetic (chemical) type, or possibly a combination of both. For practical purposes there should be no difference, since both types should essentially reflect the magnetic field existing about the time of deposition.

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7. Conclusions

1. Red limestones and maroon radiolarites, of Lower to Middle Jurassic age, from the northern calcareous Alps, have been analyzed for remanent magnetism. The rocks are exceedingly fine-grained, and the magnetic mineral present is haematite.

2. The magnetic vectors obtained are comparatively consistent, and show no reversals. A lower Jurassic (Pliensbachian) pole is suggested at latitude 53° N, longitude 112° E (east of Lake Baikal), and a Middle Jurassic pole at latitude 58°·5 N and longitude 128° E (in the Aldan region, south of Yakutsk).

3. The Mid-Jurassic position agrees remarkably well with a Jurassic pole position predicted by Irving, based on extrapolation from British data on Triassic and Eocene pole positions.

4. The results indicate that very fine-grained haematite-bearing magnetite-free sediments may be more reliable palaeomagnetic indicators than the typical red siltstone or lava.

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